

GRASKOP, South Africa

WMO: 682870

Lat: 24.935S

Lon: 30.838E

Elev: 1436

StdP: 85.22

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.7	4.8	-8.6	2.2	14.2	-5.7	2.8	14.3	6.9	18.2	5.8	14.7	1.5	40	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	28.2	17.1	26.8	17.4	25.6	17.5	21.1	24.5	20.4	23.8	19.8	23.1	3.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	17.7	22.7	19.4	16.8	22.0	18.7	16.1	21.4	68.5	24.6	66.0	23.8	63.6	23.1	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.1	4.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	18.7	19.0	18.1	16.3	14.5	12.7	11.6	13.9	16.0	16.2	17.4	18.5
	DBStd	3.71	1.99	2.03	2.30	2.45	2.55	2.44	2.80	3.38	3.82	3.64	3.42	2.88
	HDD10.0	29	0	0	0	0	2	3	15	6	2	1	1	0
	HDD18.3	1042	20	15	31	68	119	168	209	143	88	88	56	35
	CDD10.0	2242	269	252	252	190	142	85	64	126	183	193	224	262
	CDD18.3	213	31	34	24	8	2	0	1	4	19	21	29	39
	CDH23.3	1022	103	88	79	33	11	2	1	35	153	168	182	166
	CDH26.7	143	10	6	7	1	0	0	0	2	27	32	32	25
Wind	WSAvg	2.1	2.0	2.0	2.1	1.9	2.0	2.0	2.1	2.5	2.3	2.3	2.4	2.1
Precipitation	PrecAvg	1205	208	182	156	69	22	13	14	17	34	76	147	204
	PrecMax	2274	402	713	347	172	110	67	47	102	148	143	377	430
	PrecMin	521	59	10	29	1	0	0	0	0	13	46	65	65
	PrecStd	365	96	164	89	40	25	17	13	21	33	35	68	82
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.4	27.4	27.5	26.2	24.7	23.3	22.5	26.5	29.5	29.7	29.5	29.2
		MCWB	19.5	20.2	18.8	16.9	14.5	12.0	10.0	12.9	13.5	16.4	17.7	19.9
	2%	DB	25.7	25.6	25.4	24.2	22.8	21.1	20.2	24.3	27.1	27.1	27.1	26.8
		MCWB	19.7	19.5	18.5	16.5	13.8	11.4	9.7	12.3	13.5	16.7	17.9	19.6
	5%	DB	24.3	24.2	23.8	22.2	21.2	19.5	18.4	22.2	25.0	25.1	25.3	25.1
		MCWB	19.3	19.0	18.4	16.6	13.5	10.7	9.4	11.7	13.7	16.2	17.7	19.2
	10%	DB	22.9	23.0	22.4	20.6	19.6	17.8	16.5	20.2	22.8	22.9	23.5	23.5
		MCWB	18.9	18.7	18.1	16.2	13.2	10.3	9.3	11.3	13.8	15.7	17.3	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	21.5	21.4	20.0	17.2	14.1	13.1	15.6	17.9	19.8	20.7	21.9
		MCDB	24.5	24.9	24.3	23.0	20.8	19.6	16.8	21.7	22.8	25.5	25.5	25.9
	2%	WB	20.7	20.5	20.2	18.5	15.8	12.8	11.9	14.4	16.6	18.4	19.5	20.7
		MCDB	23.7	24.1	23.3	21.4	19.8	18.1	15.9	19.9	22.3	23.5	24.0	24.5
	5%	WB	20.0	19.8	19.3	17.6	14.9	12.0	11.1	13.4	15.6	17.4	18.7	20.0
		MCDB	22.9	23.2	22.4	20.5	19.1	17.5	15.4	18.6	21.1	22.4	23.1	23.6
	10%	WB	19.4	19.1	18.5	16.8	14.1	11.3	10.2	12.6	14.7	16.4	17.9	19.2
		MCDB	22.1	22.2	21.4	19.5	17.8	16.1	14.5	17.5	20.1	21.2	22.2	22.9
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.4	7.5	7.7	10.1	10.5	9.6	10.4	10.5	9.3	8.7	7.5
		MCDBR	10.2	9.8	10.5	10.6	11.9	12.3	12.1	13.5	13.8	13.3	12.6	10.8
		MCWBR	5.3	4.9	5.1	5.5	5.6	5.8	5.5	5.4	5.4	5.6	5.7	5.4
	5% WB	MCDBR	8.2	8.6	9.0	8.6	10.3	10.8	10.1	10.9	11.2	10.9	10.5	9.7
		MCWBR	4.9	5.0	5.3	5.3	5.9	5.9	5.8	5.4	5.9	6.0	5.7	5.6
Clear-Sky Solar Irradiance	taub		0.366	0.360	0.355	0.340	0.309	0.293	0.294	0.346	0.395	0.408	0.379	0.379
	taud		2.429	2.449	2.452	2.452	2.500	2.524	2.508	2.350	2.240	2.250	2.369	2.384
	Ebn at Noon		977	968	944	913	903	899	911	892	889	911	958	966
	Edh at Noon		124	119	113	103	90	83	88	112	137	143	131	130
All-Sky Solar Radiation	RadAvg		6.17	6.03	5.37	4.59	4.26	3.84	4.08	4.74	5.37	5.53	5.85	6.08
	RadStd		0.45	0.51	0.42	0.33	0.19	0.23	0.14	0.27	0.33	0.37	0.50	0.46

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page